

Statistical Analysis Report

Analysis Summary Template

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1 Summary

Provide a brief overview of the analysis objectives, key findings, and main conclusions. This section should be concise and accessible to non-technical stakeholders. Max 1 paragraph, no bullet points

2 Introduction

This analysis is to demonstrate the use of Artificial Intelligence (AI) Large Language Models (LLMs) to analyse data. The goal here is to stress test LLMs on a simple analysis task that involves a linear mixed effects model with a binary dependent variable and a blocking factor. The report will be in the form of a statistical analysis report.

3 Data

The data are from a subset of the Chz2006 dataset in the R package {agricolae} (Mendiburu and Yaseen 2020). There are 3 factors of interest (crop, amendment, and block) and a binary dependent variable (wilt_percent). There is also a plant individual factor which we will not consider here. There are $n = 1920$ observations in the data. The data experimentally measures the incidence and performance of healthy tubers and rotten potato field infested with naturally *Ralstonia solanacearum* Race 3/Bv 2A, after application of inorganic amendments and a rotation crop in Carhuaz Peru, 2006.

```
'data.frame':  1920 obs. of  6 variables:
 $ amendment  : chr  "OC" "OC" "OC" "OC" ...
 $ crop        : chr  "Pea" "Pea" "Pea" "Pea" ...
```

```

$ block      : Factor w/ 3 levels "1","2","3": 1 1 1 1 1 1 1 1 1 1 1 ...
$ plant      : int   1 2 3 4 5 6 7 8 9 10 ...
$ wilt_percent: int   0 0 50 0 0 0 0 0 0 0 0 ...
$ wilt_bin   : num   0 0 1 0 0 0 0 0 0 0 0 ...

```

Table 1: Dataset sample

amendment	crop	block	plant	wilt_percent	wilt_bin
0C	Pea	1	1	0	0
0C	Pea	1	2	0	0
0C	Pea	1	3	50	1
0C	Pea	1	4	0	0
0C	Pea	1	5	0	0
0C	Pea	1	6	0	0

4 Statistical Analysis

The statistical analysis consisted of a linear mixed effects model with a binary dependent variable and a blocking factor treating block as a random effect. The model was fitted using the lme4 package in R.

5 Results

The results of the analysis are presented below with the predictor names and the corresponding p-values, accompanied by a graph of the relationship between each predictor and the response variable.

```
[1] "Model Summary"
```

```

Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: binomial ( logit )
Formula: wilt_bin ~ crop + amendment + (1 | block)
Data: wilt

```

AIC	BIC	logLik	-2*log(L)	df.resid
1146.5	1191.0	-565.3	1130.5	1912

Scaled residuals:

Min	1Q	Median	3Q	Max
-0.9244	-0.3003	-0.2264	-0.0968	7.8011

Random effects:

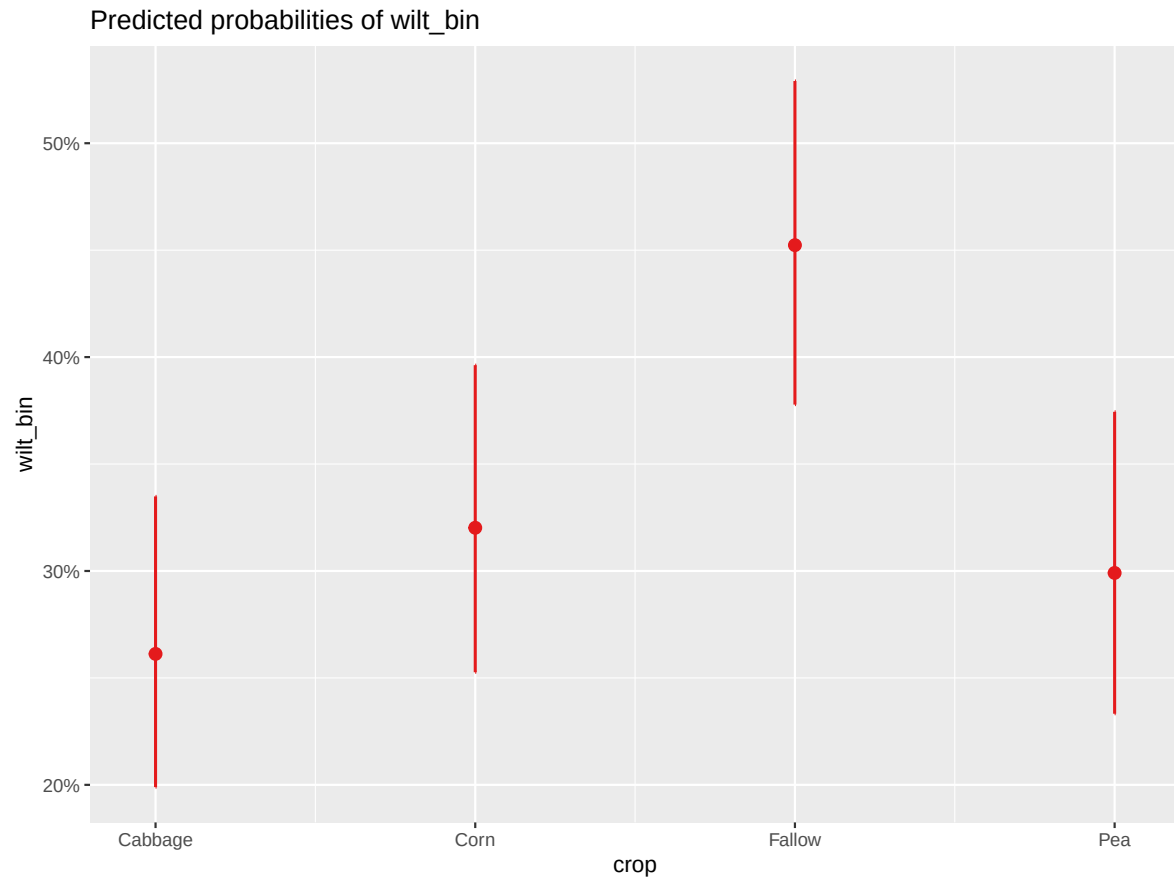
Groups Name	Variance	Std.Dev.
block (Intercept)	0.003477	0.05897

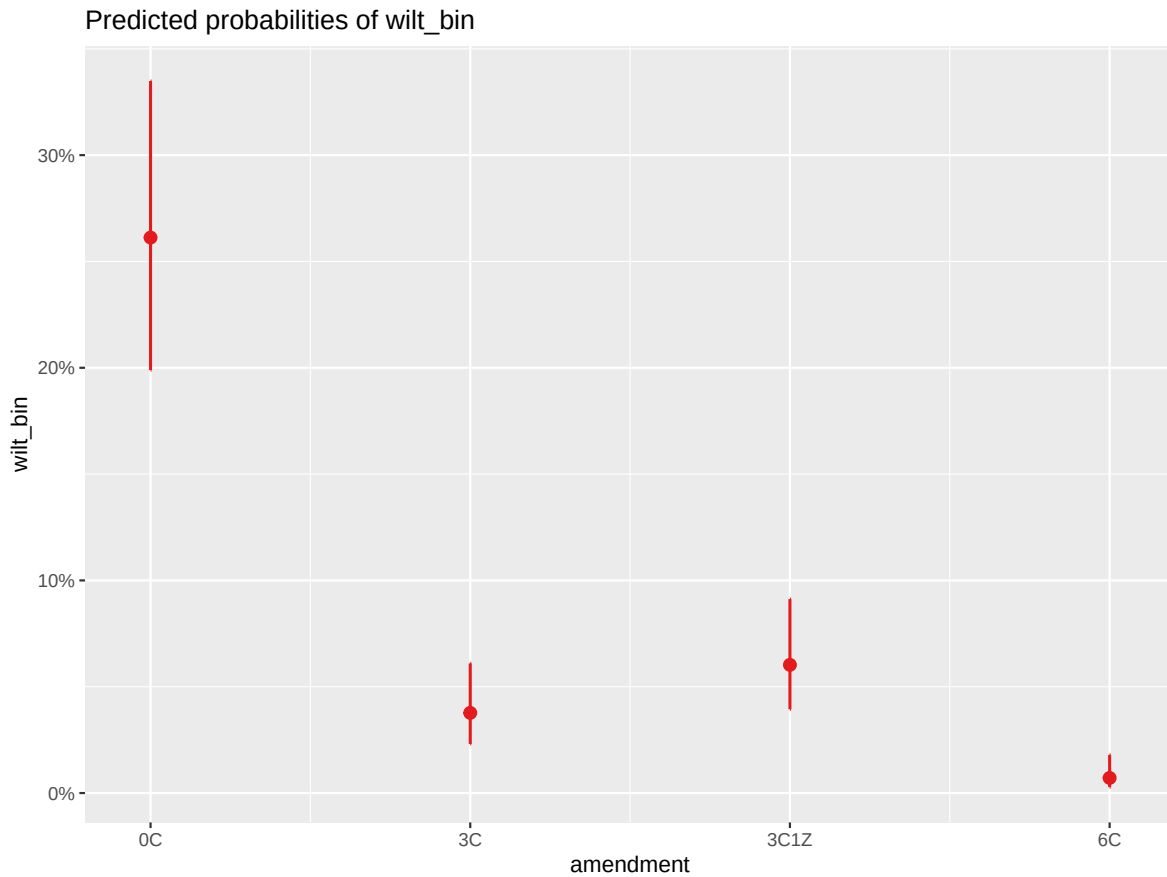
Number of obs: 1920, groups: block, 3

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.0396	0.1803	-5.766	8.11e-09 ***
cropCorn	0.2867	0.2291	1.252	0.211
cropFallow	0.8484	0.2170	3.909	9.28e-05 ***
cropPea	0.1879	0.2320	0.810	0.418
amendment3C	-2.1997	0.2254	-9.761	< 2e-16 ***
amendment3C1Z	-1.7068	0.1917	-8.905	< 2e-16 ***
amendment6C	-3.8991	0.4608	-8.462	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1





6 References

Mendiburu, Felipe de and Yaseen, Muhammad(2020). agricolae: Statistical Procedures for Agricultural Research.R package version 1.4.0, <https://myaseen208.github.io/agricolae/><https://cran.r-project.org/package=agricolae>.

7 Appendix

Session Information

```
sessionInfo()
```

```
R version 4.5.1 (2025-06-13 ucrt)
Platform: x86_64-w64-mingw32/x64
```

Running under: Windows 11 x64 (build 26100)

Matrix products: default

LAPACK version 3.12.1

locale:

```
[1] LC_COLLATE=English_United Kingdom.utf8
[2] LC_CTYPE=English_United Kingdom.utf8
[3] LC_MONETARY=English_United Kingdom.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United Kingdom.utf8
```

time zone: Europe/London

tzcode source: internal

attached base packages:

```
[1] stats      graphics  grDevices  utils      datasets  methods    base
```

other attached packages:

```
[1] sjPlot_2.9.0      lmerTest_3.1-3    lme4_1.1-37       Matrix_1.7-3
[5] knitr_1.50        lubridate_1.9.4   forcats_1.0.1     stringr_1.5.1
[9] dplyr_1.1.4       purrr_1.0.4       readr_2.1.5       tidyr_1.3.1
[13] tibble_3.3.0      ggplot2_3.5.2     tidyverse_2.0.0
```

loaded via a namespace (and not attached):

```
[1] generics_0.1.4      stringi_1.8.7      lattice_0.22-7
[4] hms_1.1.3           digest_0.6.37      magrittr_2.0.3
[7] evaluate_1.0.4       grid_4.5.1         timechange_0.3.0
[10] RColorBrewer_1.1-3  fastmap_1.2.0      jsonlite_2.0.0
[13] ggeffects_2.3.1     scales_1.4.0       numDeriv_2016.8-1.1
[16] reformulas_0.4.1    Rdpack_2.6.4       cli_3.6.5
[19] rlang_1.1.6         rbibutils_2.3      splines_4.5.1
[22] withr_3.0.2         yaml_2.3.10        datawizard_1.3.0
[25] tools_4.5.1         tzdb_0.5.0         nloptr_2.2.1
[28] minqa_1.2.8         sjlabelled_1.2.0   boot_1.3-31
[31] vctr_0.6.5          R6_2.6.1           lifecycle_1.0.4
[34] MASS_7.3-65         insight_1.4.2      pkgconfig_2.0.3
[37] pillar_1.10.2       gtable_0.3.6       glue_1.8.0
[40] Rcpp_1.0.14         haven_2.5.5        xfun_0.52
[43] tidyselect_1.2.1    farver_2.1.2       htmltools_0.5.8.1
[46] nlme_3.1-168        labeling_0.4.3     rmarkdown_2.29
[49] compiler_4.5.1
```